QCM A · antiderivative families

After S53; QCM B after S58.

For each question, only one proposal is accurate.
consult the correction.

1. A antiderivative of x^2 on $\mathbb{R}$ is:

A. $2x$

B. x^3

C. $x^3/3$

D. $3x^3$

2. All antiderivatives of e^{2x} are:

A. $e^{2x} + C$

B. $\frac{1}{2}e^{2x} + C$

C. $2e^{2x} + C$

D. $xe^{2x} + C$

3. A antiderivative of $2x/(1+x^2)$ on $\mathbb{R}$ is:

A. $\frac{1}{1+x^2}$

B. $2\ln x$

C. $\ln x$

D. $\ln(1+x^2)$

4. Two antiderivatives of the same function over an interval have curves:

A. Images of one another by vertical translation

B. Necessarily identical

C. Always stinging

D. Images by any horizontal translation

5. The antiderivative of $2x$ verifier $F(1) = 3$ is:

A. $x^2 + 3$ B. $2x^2 + 1$

C. $x^2 + 2$ D. $x^2 - 2$

QCM B · Differential Equation and Model

For each question, only one proposal is accurate.
consult the correction.

6. The solution of $y' = -2y$, $y(0) = 3$, is:

A. $3e^{2t}$ B. $3e^{-2t}$

C. $-2e^{3t}$ D. $3 - 2t$

7. The constant solution of $y' = -2y + 6$ is:

A. 6 B. -3

C. 2 D. 3

8. If y_p is a solution of $y' = ay + b(t)$, then $z = y - y_p$ verify:

A. $z' = az$ B. $z' = az + b(t)$

C. $z' = b(t)$ D. $z' = 0$ necessarily

9. Step by step h for $y' = -y$, from $y_0 = 2$, gives:

A. $y_1 = 2 + h$ B. $y_1 = e^{-h}$

C. $y_1 = 2 - 2h$ D. $y_1 = 2 - h^2$

10. If t is in minutes and T in degrees Celsius, the coefficient k of $T' = -k(T - 20)$ expressed in:

A. minutes B. Inverse minutes

C. degrees Celsius D. litres

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. A antiderivative is unique without additional conditions.

2. Euler gives the exact value for any differential equation.

3. Checking a solution consists of checking the equation and the initial condition.